

Tea Blending Optimisation: A Comprehensive Analysis

STA5074Z Prescriptive Analytics

Tinotenda Muzambi - MZMTIN002

2025-11-16

Contents

1 Introduction	2
2 Problem Formulation and Data Preparation	2
2.1 Notations and Parameters	2
2.2 Data Source	2
3 Data Implementation (Fomeni 2018)	3
3.1 Data Loading	3
3.2 Data Characteristics	4
4 Approach 1: Cost Minimisation using Linear Programming	7
4.1 Model Formulation	7
4.2 Implementation	8
4.3 Visualisation	9
5 Approach 2: Cost Minimisation using Simulated Annealing	9
5.1 Implementation	9
5.2 Visualisation	10
6 Approach 3: Multi-objective Optimisation using Chebyshev Goal Programming	11
6.1 Model Formulation	11
6.2 Implementation	12
6.3 Visualisation	13
7 Approach 4: Genetic Algorithms	13
7.1 Implementation	13
7.2 Visualisation	14
8 Summary of Results and Comparative Analysis	15
8.1 Comparative Metrics	15
8.2 Key Findings	16
8.3 Solution Matrices Comparison	17
9 Conclusion	18
10 Appendix: Code Execution Notes	19
10.1 Required Libraries	19
10.2 Reproducibility	19
10.3 Parameter Tuning	19

11 Appendix: Complete Recipe Matrices	19
11.1 Linear Programming Solution	19
11.2 Simulated Annealing Solution	21
11.3 Chebyshev Goal Programming Solution	23
11.4 Genetic Algorithm Solution	25
References	27

1 Introduction

Blending problems are a classic challenge in operations research, arising in diverse fields such as food processing, metallurgy, and energy. In the tea industry, blending is crucial to achieve consistent product quality, balance raw material costs, and adapt to fluctuating supplies. Every cup of tea on the supermarket shelf is the result of a carefully orchestrated process. Different grades and types of tea leaves from various origins are combined to meet precise targets for taste, colour, aroma and other quality parameters, all while minimising production costs and ensuring supply constraints are not violated.

In the context of tea manufacturing, the blending optimisation problem involves selecting the optimal composition of raw teas to satisfy desired blend qualities for multiple final blends. This not only impacts the sensory profile and brand identity of each product, but also has substantial operational and economic implications. By applying advanced computational techniques, such as mathematical programming and metaheuristics, producers can automate and enhance the blending decision-making process, yielding better margins, consistent quality and greater responsiveness to market demands.

Below, we formally define the tea blending optimisation problem and systematically explore several approaches for solving it. The problem requires determining the least-cost combination of raw materials to produce final blends that meet specific quality characteristics, as formalised by Fomeni^[1].

The primary objectives are:

1. To implement and solve a cost-minimisation model using Linear Programming (LP).
2. To solve the same problem using Simulated Annealing (SA) metaheuristic and compare performance.
3. To formulate and solve a multi-objective model using Chebyshev Goal Programming.
4. To explore Genetic Algorithms (GAs) as an alternative solution approach.

2 Problem Formulation and Data Preparation

2.1 Notations and Parameters

- **Indices:**
 - i : Index for raw materials
 - j : Index for final blends
 - k : Index for quality characteristics
- **Parameters:**
 - P_i : Unit cost of raw material i (£/Kg)
 - D_j : Weekly demand for final blend j (Kgs)
 - a_i : Weekly availability of raw material i (Kgs)
 - g_{ik} : Grade of characteristic k in one unit of raw material i
 - s_{jk} : Required target score of characteristic k in one unit of final blend j
- **Decision Variables:**
 - x_{ij} : Amount (in Kgs) of raw material i to be used in final blend j

2.2 Data Source

This project uses the real-world data from Fomeni^[1], extracted from tables 2 and 3 of the paper. These data represent actual tea blending scenarios from a UK-based tea company. The data are loaded and prepared in

section 3 below.

3 Data Implementation (Fomeni 2018)

This section implements the tea blending optimisation problem using the actual data provided in Fomeni^[1]. The data consist of 60 raw materials and 10 final blends, each with 6 quality characteristics.

3.1 Data Loading

```
## === Data Summary ===
## Number of raw materials: 60
## Number of blends: 10
## Number of quality characteristics: 6
## Price range: £0.0120 - £0.0470 per kg
## === Raw Materials Data (Fomeni Table 2 - First 10 rows) ===
```

Table 1: Raw Materials: Costs (£/kg), Availability (kg), and Quality Characteristics (First 10)

Material_ID	Cost	Availability	C1	C2	C3	C4	C5	C6
R1	0.0240	5,000	5	4	9	6	5	5
R2	0.0167	1,000	5	3	9	4	5	1
R3	0.0235	25,000	5	4	9	6	5	1
R4	0.0170	1,000	5	3	9	4	5	1
R5	0.0230	25,000	5	4	9	6	5	1
R6	0.0265	20,000	7	4	9	6	7	1
R7	0.0265	20,000	7	4	9	6	7	1
R8	0.0294	1,000	8	4	9	6	8	5
R9	0.0284	15,000	8	4	9	6	8	1
R10	0.0284	15,000	8	4	9	6	8	1

```
## === Blends Data (Fomeni Table 3) ===
```

Table 2: Blends: Demand (kg) and Target Quality Scores

Blend_ID	Demand	C1	C2	C3	C4	C5	C6
B1	100,000	2.3	3.0	3.0	2.0	2.32	1
B2	100,000	3.8	4.7	2.9	3.0	3.80	1
B3	60,000	6.6	5.2	6.1	6.9	6.60	1
B4	100,000	6.2	5.2	5.3	6.4	6.20	1
B5	100,000	4.7	5.1	4.5	5.0	4.70	1
B6	50,000	8.4	4.7	6.7	7.3	8.40	1

Table 2: Blends: Demand (kg) and Target Quality Scores

Blend_ID	Demand	C1	C2	C3	C4	C5	C6
B7	10,000	4.0	0.5	1.0	5.0	4.00	1
B8	5,000	5.9	0.5	1.0	3.9	5.90	1
B9	10,000	7.0	3.9	9.0	5.9	7.00	1
B10	20,000	4.0	2.0	3.0	2.0	4.00	9

```
## === Feasibility Check ===
## Total availability: 830500 kg
## Total demand: 555000 kg
## Surplus availability: 275500 kg (49.64%)
```

3.2 Data Characteristics

```
## === Summary Statistics ===
## Raw Material Prices (£/kg):
```

Table 3: Price Distribution Statistics

Statistic	Value
Mean	£0.0217
Median	£0.0195
Minimum	£0.0120
Maximum	£0.0470
Standard Deviation	£0.0076

```
## Raw Material Availability:
```

Table 4: Availability Distribution Statistics (kg)

Statistic	Value_kg
Mean	13842
Median	5000
Minimum	500
Maximum	80000
Total	830500

```
## Quality Characteristics:
```

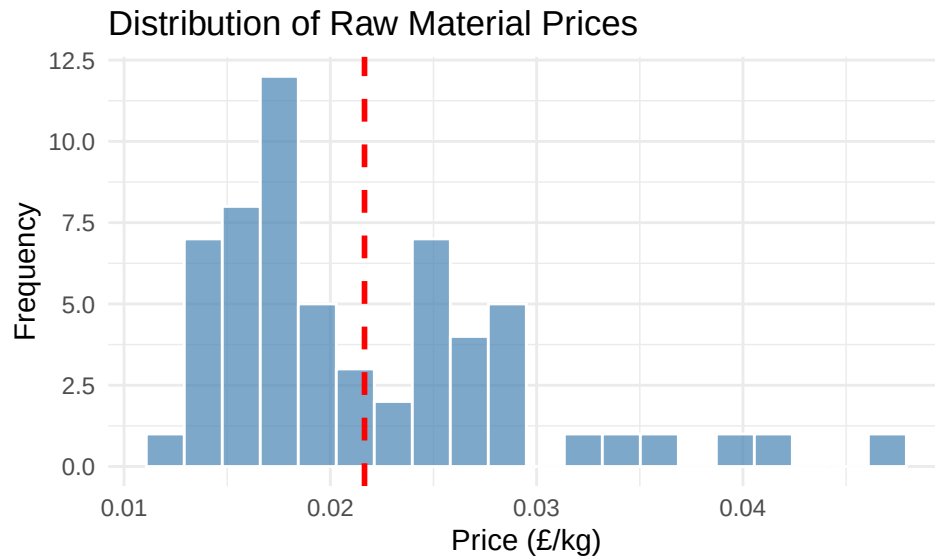
Table 5: Quality Characteristics Summary Statistics

Characteristic	Minimum	Maximum	Mean	Median	Std_Dev
C1	2.0	9	5.500000	5.0	1.917802
C2	0.5	8	3.333333	3.0	1.990798
C3	1.0	9	4.316667	4.0	2.507559
C4	2.0	8	4.500000	3.5	2.452947
C5	2.0	9	5.500000	5.0	1.917802
C6	1.0	9	1.933333	1.0	1.858254

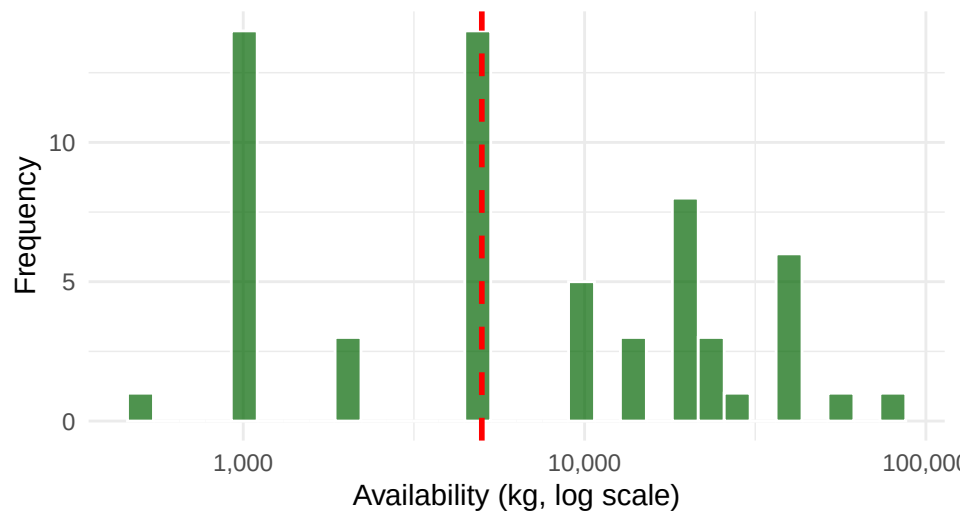
Blend Demands:

Table 6: Blend Demand Statistics (kg)

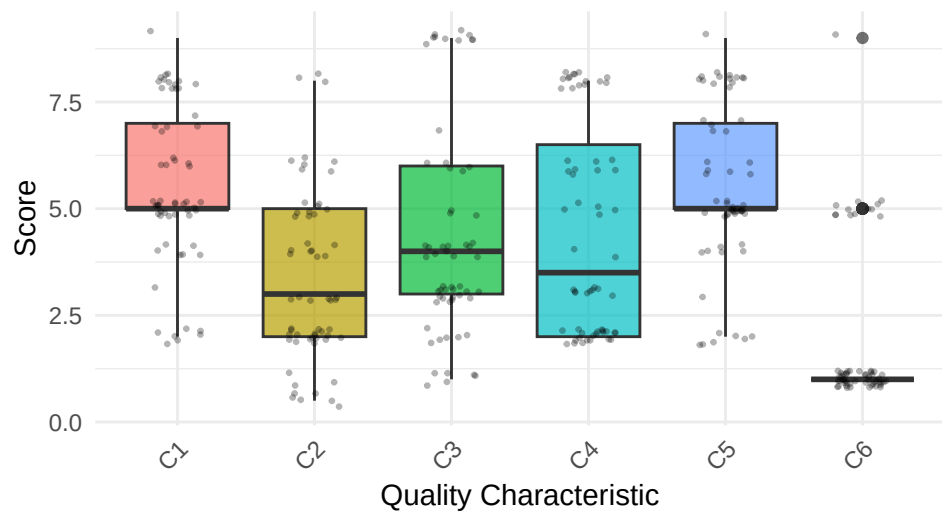
Statistic	Value_kg
Mean	55500
Median	55000
Minimum	5000
Maximum	100000
Total	555000



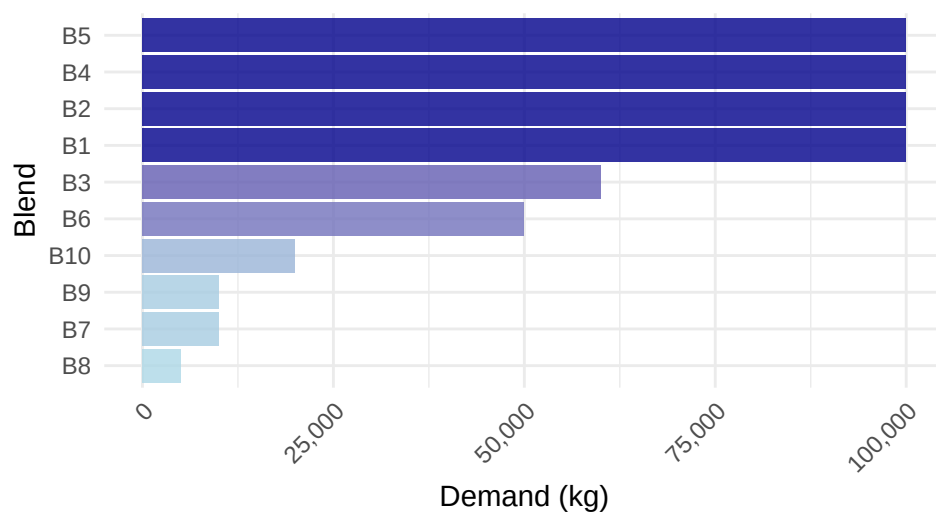
Distribution of Raw Material Availability

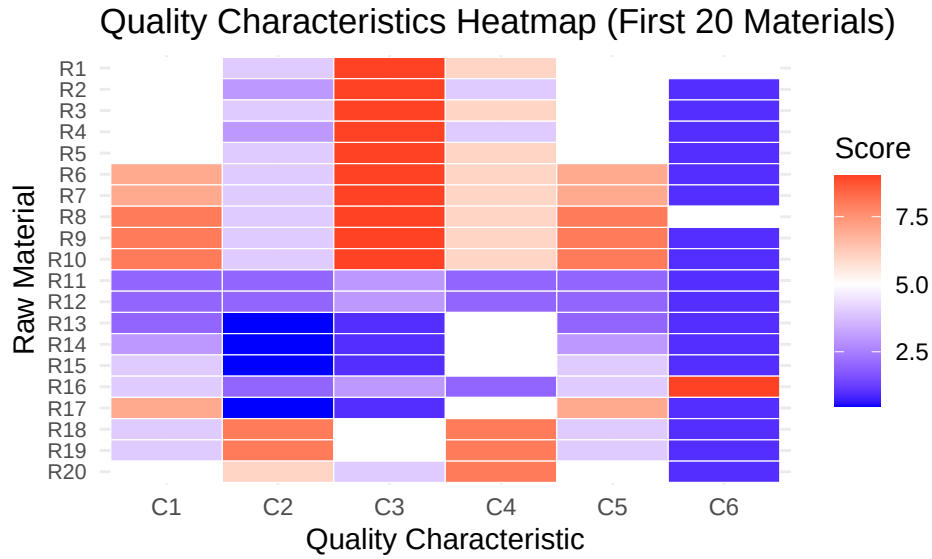
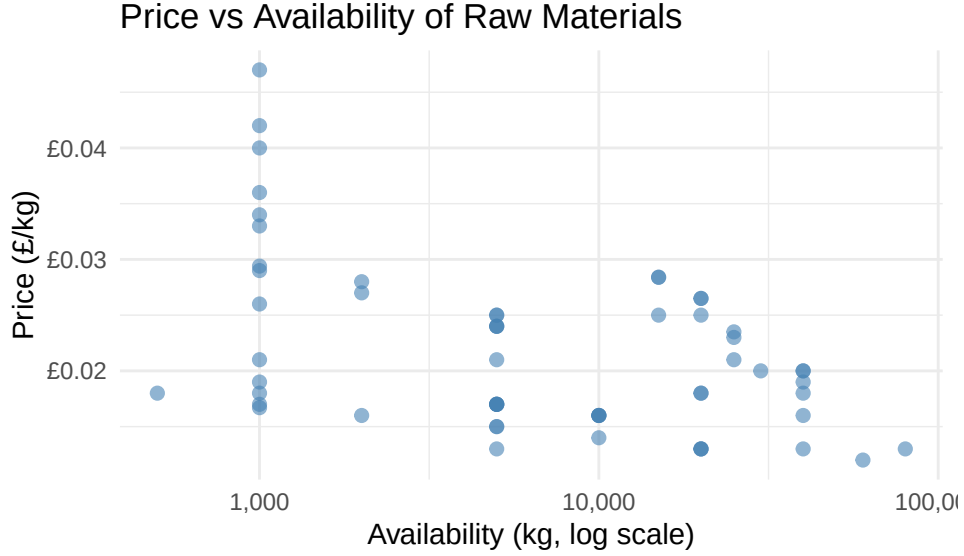


Distribution of Quality Characteristics Across Raw Materials



Demand by Blend





4 Approach 1: Cost Minimisation using Linear Programming

This section implements the cost-minimisation model using Linear Programming with parametric relaxation of the characteristic-matching constraints. This involves implementing a cost-minimisation LP model that finds the optimal combination of raw materials to produce blends meeting demand and quality constraints. This implementation uses Rglpk which provides a straightforward matrix-based formulation that is well-suited for this problem type.

4.1 Model Formulation

Objective Function: Minimise total cost

$$\min \sum_i \sum_j P_i x_{ij}$$

Subject to Constraints:

1. Raw Material Availability: $\sum_j x_{ij} \leq a_i \quad \forall i$

2. Blend Demand: $\sum_i x_{ij} \geq D_j \quad \forall j$
3. Characteristic Score Relaxation:
 - $\sum_i (g_{ik} - (s_{jk} + \epsilon_{jk}))x_{ij} \leq 0 \quad \forall j, k$
 - $\sum_i (g_{ik} - (s_{jk} - \epsilon_{jk}))x_{ij} \geq 0 \quad \forall j, k$
4. Non-negativity: $x_{ij} \geq 0 \quad \forall i, j$

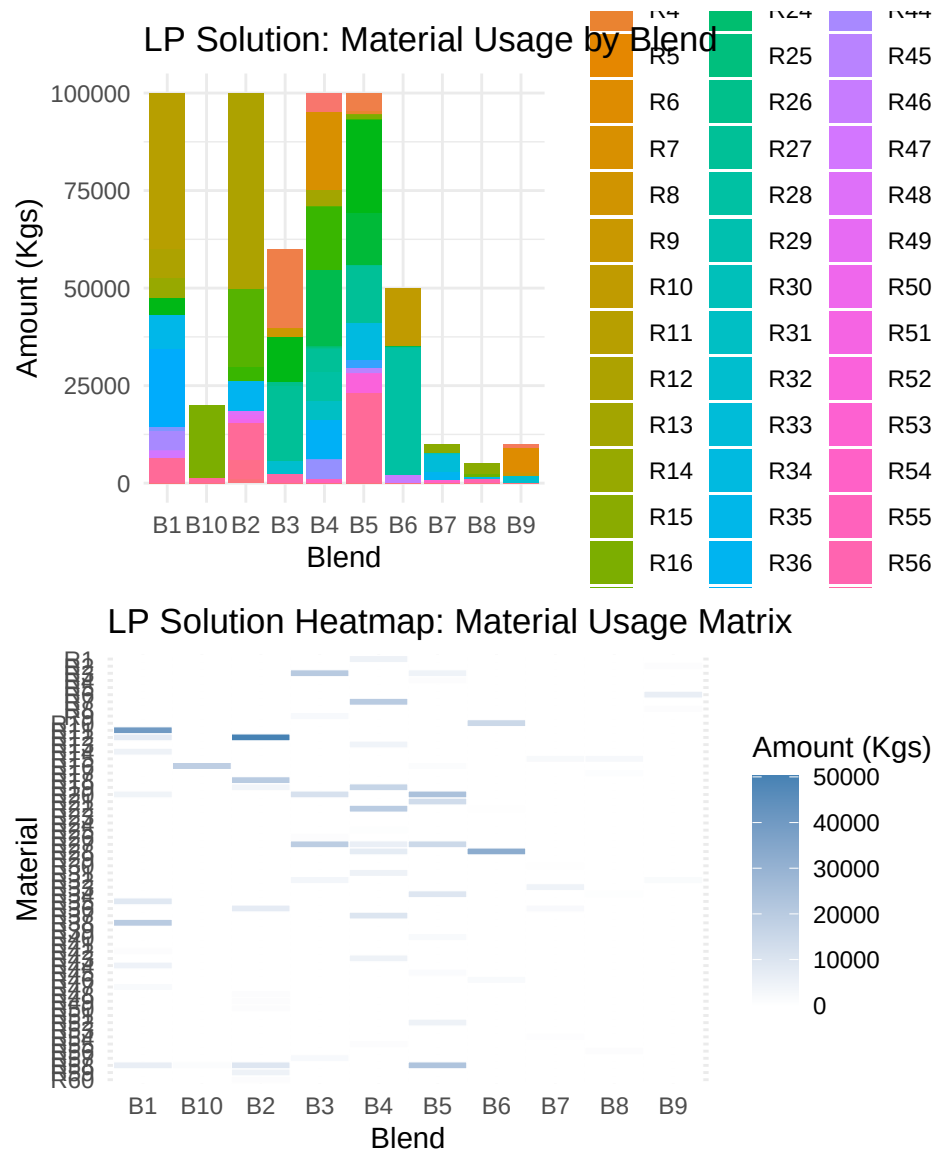
4.2 Implementation

```
## === LP Solution Results ===
## Solver Status: 0 (0 = optimal)
## Optimal Total Cost: £9087.69
## Total material used: 555000.00 Kgs
## Total demand required: 555000.00 Kgs
## === LP Solution Summary ===
## Total materials used: 52 out of 60
## Materials with non-zero allocation: R1, R2, R3, R4, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16
## Note: The complete recipe matrix (all 60 materials × 10 blends) is provided in Appendix 11.1.
## === Quality Deviations from Targets ===
```

Table 7: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	0.5000000	0.5000000	0.05429705	0.5000000	0.4800000	0.19873016
B2	0.5000000	0.5000000	0.5000000	0.5000000	0.5000000	0.12126984
B3	0.3736758	0.5000000	0.5000000	0.1152129	0.3736758	0.06666667
B4	0.5000000	0.43809524	0.24592593	0.1150794	0.5000000	0.2400000
B5	0.5000000	0.32182063	0.5000000	0.5000000	0.5000000	0.1000000
B6	0.5000000	0.07333333	0.10666667	0.1400000	0.5000000	0.1600000
B7	0.3977401	0.5000000	0.5000000	0.5000000	0.3977401	0.23728814
B8	0.5000000	0.2500000	0.5000000	0.3000000	0.5000000	0.0000000
B9	0.2333333	0.16666667	0.5000000	0.2333333	0.2333333	0.4000000
B10	0.0000000	0.3750000	0.0000000	0.0000000	0.0000000	0.5000000

4.3 Visualisation



5 Approach 2: Cost Minimisation using Simulated Annealing

Simulated Annealing is a metaheuristic that explores the solution space probabilistically, accepting worse solutions with decreasing probability as the algorithm “cools down.”

5.1 Implementation

```
## Running Simulated Annealing...
## === SA Solution Results ===
## Best Total Cost: £10777.97
## Difference from LP: 18.60%
## === SA Solution Summary ===
## Total materials used: 60 out of 60
```

Materials with non-zero allocation: R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15

Note: The complete recipe matrix (all 60 materials $\times$ 10 blends) is provided in Appendix 11.2.

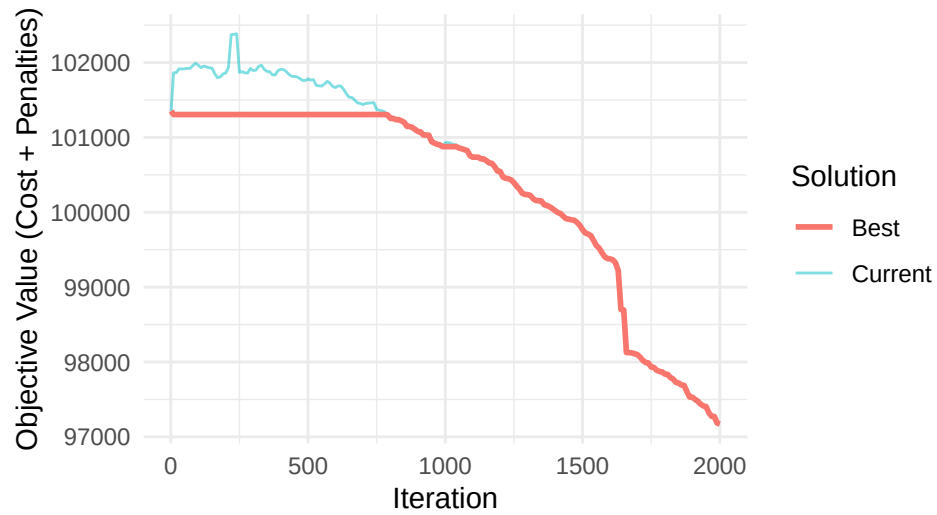
=== Quality Deviations from Targets ===

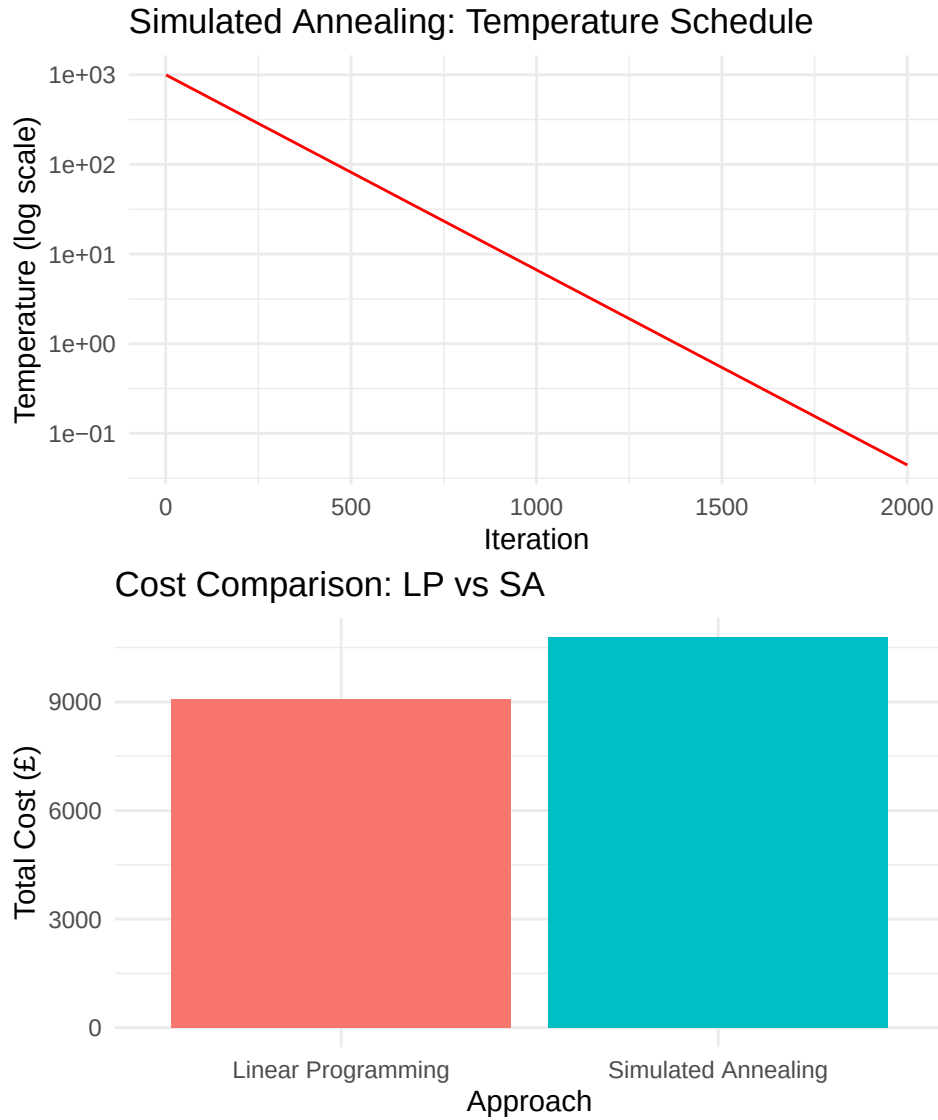
Table 8: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	2.7100203	0.4865980	1.6845582	3.0594142	2.6900203	0.5452768
B2	1.4330087	0.6512513	1.7941613	2.5284218	1.4330087	0.3973148
B3	1.2382151	1.3462221	1.0849102	1.3160990	1.2382151	0.4048457
B4	0.9904153	1.4561789	0.8412904	1.1878815	0.9904153	0.3549188
B5	0.6338601	1.2201161	0.4954091	0.1665509	0.6338601	0.3412532
B6	2.9987062	0.5694051	2.2106917	2.0053458	2.9987062	0.3015301
B7	1.3829100	3.0007512	3.8371827	0.4274194	1.3829100	0.2370485
B8	0.6653449	3.1714699	3.4317411	1.0092582	0.6653449	0.4911140
B9	1.5891256	0.2883281	4.4640362	0.7513100	1.5891256	0.2269901
B10	1.1774699	1.6284395	1.5010893	3.0893268	1.1774699	7.3903869

5.2 Visualisation

Simulated Annealing: Convergence Plot





6 Approach 3: Multi-objective Optimisation using Chebyshev Goal Programming

Chebyshev Goal Programming minimises the worst-case deviation from multiple goals, providing a balanced solution that trades off cost for quality control.

6.1 Model Formulation

Objective Function: Minimise worst-case deviation

$$\min \max\{w_0 \cdot d_0, w_{jk} \cdot d_{jk}\}$$

Where: - d_0 : Deviation from cost goal - d_{jk} : Deviation from quality goal for blend j , characteristic k

Constraints: - All LP constraints - Goal constraints for cost and quality

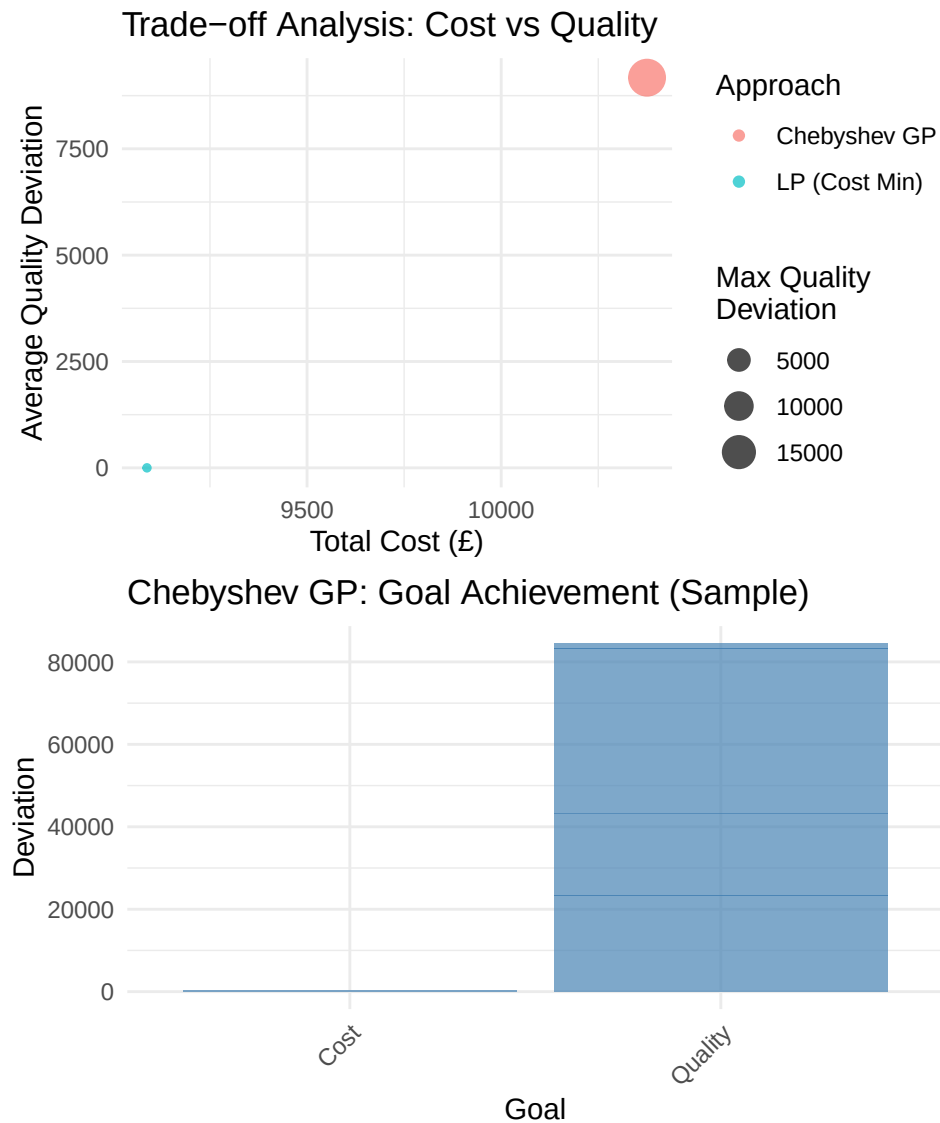
6.2 Implementation

```
## Solving Chebyshev Goal Programming model...
## === Chebyshev GP Solution Results ===
## Worst-case Deviation (z): 39945.0309
## Total Cost: £10375.59
## Cost Deviation from Target: £379.13
## === Chebyshev GP Solution Summary ===
## Total materials used: 48 out of 60
## Materials with non-zero allocation: R2, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R16, R17, R
## Note: The complete recipe matrix (all 60 materials × 10 blends) is provided in Appendix 11.3.
## === Quality Deviations from Targets ===
```

Table 9: Quality deviations (minimised in worst-case sense)

Blend	C1	C2	C3	C4	C5	C6
B1	3,324.17181172407435952	19,972.515	6,657.505	19,972.515	5,324.17181172407345002	0.00
B2	19,972.51543517221944057	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B3	0.00000000002763036	19,972.515	0.000	16,297.229	0.00000000002763036	0.00
B4	19,972.51543517226309632	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B5	19,972.51543517226673430	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B6	19,972.51543517226309632	1,761.103	16,742.083	19,972.515	19,972.51543517226673430	4,000.00
B7	0.00000000000000000	0.000	0.000	0.000	0.00000000000000000	0.00
B8	0.00000000000000000	0.000	0.000	5,500.000	0.00000000000000000	0.00
B9	1,250.00000000000000000	500.000	1,000.000	0.000	1,250.00000000000000000	0.00
B10	0.00000000000000000	0.000	0.000	821.115	0.00000000000000000	6,568.92

6.3 Visualisation



7 Approach 4: Genetic Algorithms

Genetic Algorithms use evolutionary principles to evolve a population of solutions toward better fitness.

7.1 Implementation

```
## Running Genetic Algorithm...
```

```
## === GA Solution Results ===
```

```
## Best Total Cost: £15396.39
```

```
## Average Quality Deviation: 1.4674
```

```
## === GA Solution Summary ===
```

```
## Total materials used: 60 out of 60
```

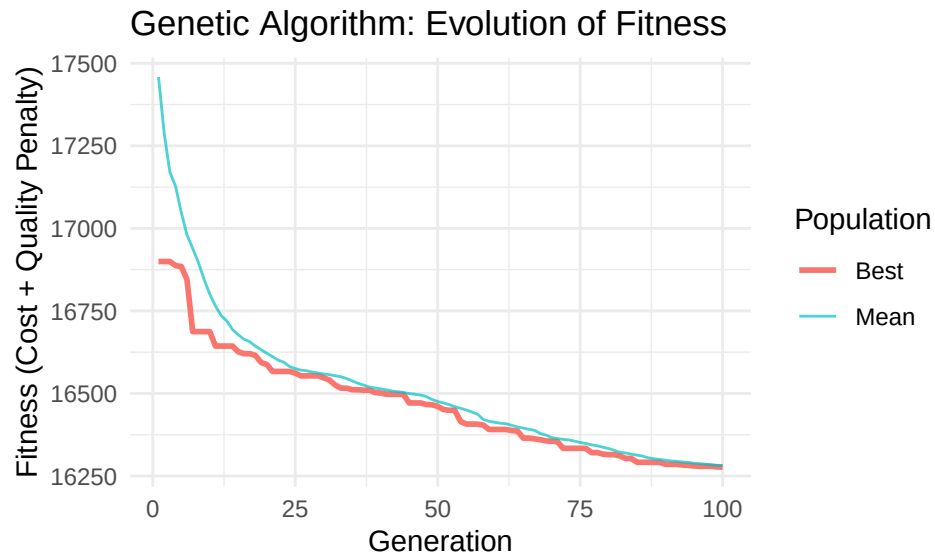
```
## Materials with non-zero allocation: R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15
```

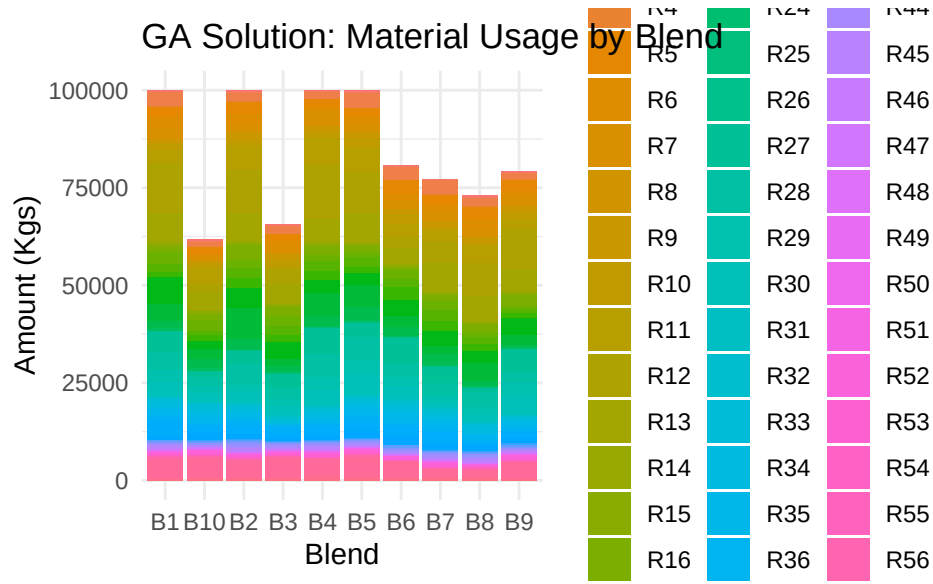
Note: The complete recipe matrix (all 60 materials × 10 blends) is provided in Appendix 11.4.
 ## === Quality Deviations from Targets ===

Table 10: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	2.5116737	0.6917901	1.40307406	3.1379566	2.4916737	0.2056858
B2	0.8728974	0.9411929	1.40994714	2.0893307	0.8728974	0.3935503
B3	1.4825316	1.1838712	1.57926383	1.5844308	1.4825316	0.4324463
B4	1.3685800	1.4236709	0.91133388	1.2872199	1.3685800	0.2547486
B5	0.1806112	1.3235059	0.05704217	0.1500934	0.1806112	0.2052863
B6	3.1210859	0.6742477	1.92795440	1.7852562	3.1210859	0.1553911
B7	0.8415012	3.1433002	3.42807585	0.2475233	0.8415012	0.2682886
B8	1.2192344	3.0267049	3.41900339	1.1035626	1.2192344	0.3166768
B9	1.9907466	0.1910143	4.43096864	0.8136152	1.9907466	0.4074426
B10	0.8238665	1.6998966	1.01722208	2.8838559	0.8238665	7.6301806

7.2 Visualisation





8 Summary of Results and Comparative Analysis

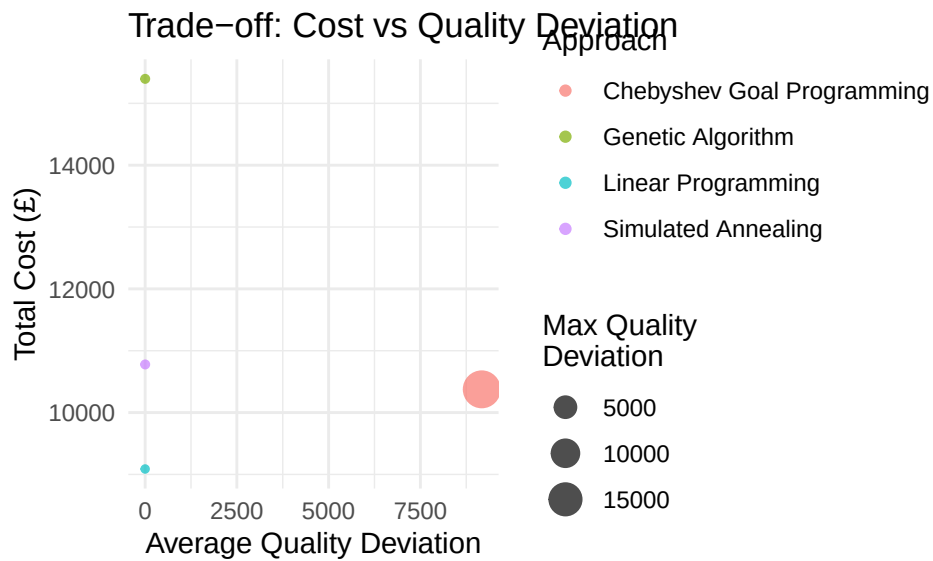
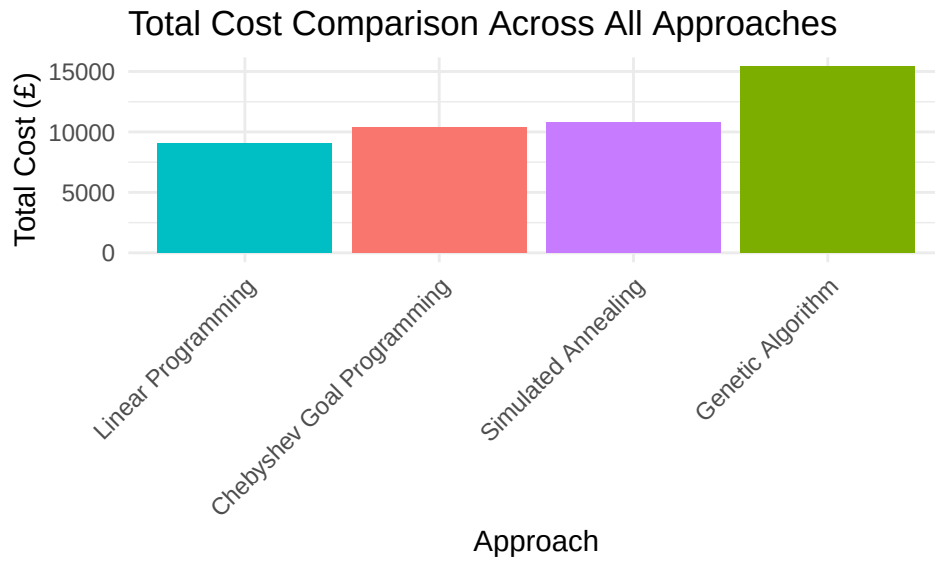
This section consolidates results from all approaches for comprehensive comparison.

8.1 Comparative Metrics

=== Comprehensive Comparison of All Approaches ===

Table 11: Performance Metrics Comparison

Approach	Total_Cost	Avg_Quality_Dev	Max_Quality_Dev	Cost_vs_LP_pct
Linear Programming	9,087.691	0.3324814	0.500000	0.00000
Simulated Annealing	10,777.974	1.5168948	7.390387	18.59970
Chebyshev Goal Programming	10,375.592	9,172.2778281	19,972.515435	14.17193
Genetic Algorithm	15,396.386	1.4673513	7.630181	69.42023



8.2 Key Findings

=== Key Findings ===

1. Cost Performance:

- LP provides the optimal cost baseline: £9087.69

- SA achieves 118.60% of LP cost

- Chebyshev GP trades 14.17% cost increase for better quality control

- GA achieves 169.42% of LP cost

2. Quality Performance:

- Best average quality: Linear Programming (deviation: 0.3325)

3. Recommendation:

- Linear Programming provides the best cost-performance trade-off.

- For near-optimal solutions quickly, Simulated Annealing is valuable.

8.3 Solution Matrices Comparison

=== Solution Matrix Comparison (Sample: Blend 1) ===

Table 12: Material allocation for Blend 1 across all approaches

Material	LP	SA Chebyshev_GP	GA
R1	0.000 1,331.85523	0.000	545.81891
R2	0.000 118.25196	0.000	57.61176
R3	0.000 3,229.15059	0.000	3,621.03608
R4	0.000 265.63224	0.000	89.17773
R5	0.000 4,171.30868	0.000	2,626.83581
R6	0.000 5,109.09689	0.000	2,568.15947
R7	0.000 4,888.28021	0.000	3,052.05712
R8	0.000 70.95111	0.000	75.81071
R9	0.000 1,930.64596	0.000	1,265.52356
R10	0.000 2,153.39220	0.000	1,517.95210
R11	40,000.000 828.26122	40,000.000	3,334.23571
R12	7,554.827 7,108.47289	32,528.357	12,743.72460
R13	0.000 5,408.09618	0.000	7,195.17407
R14	5,000.000 1,153.62356	0.000	685.94362
R15	0.000 520.77789	0.000	741.35734
R16	0.000 4,844.67525	0.000	1,504.90084
R17	0.000 2,747.51850	0.000	2,902.41474
R18	0.000 4,776.27980	3,328.753	2,121.22076
R19	0.000 84.78120	0.000	1,240.75570
R20	4,371.720 2,435.43507	0.000	6,801.31444
R21	0.000 5,018.19509	0.000	4,616.45083
R22	0.000 1,995.82237	0.000	1,645.34408
R23	0.000 1,405.25696	0.000	757.77331
R24	0.000 232.96545	0.000	100.45735
R25	0.000 57.22893	0.000	23.92929
R26	0.000 126.97315	0.000	91.69086
R27	0.000 5,101.26631	0.000	5,087.12299
R28	0.000 1,193.48229	0.000	4,829.44380
R29	0.000 1,925.78794	0.000	3,103.53837
R30	0.000 3,051.27762	0.000	3,349.97226
R31	0.000 71.74027	0.000	491.01632

Table 12: Material allocation for Blend 1 across all approaches

Material	LP	SA	Chebyshev_GP	GA
R32	0.000	325.98550	0.000	436.23961
R33	0.000	2,597.58688	0.000	1,098.96626
R34	0.000	39.74955	0.000	1,224.76929
R35	8,769.679	2,339.90929	0.000	1,033.00212
R36	0.000	1,222.14316	0.000	1,207.41370
R37	0.000	1,939.95505	0.000	1,301.90768
R38	20,000.000	4,507.38889	14,133.729	3,549.94343
R39	0.000	813.50522	0.000	949.95851
R40	0.000	272.06474	0.000	168.33999
R41	0.000	942.28110	0.000	432.53532
R42	1,000.000	370.95680	0.000	164.36775
R43	0.000	743.75571	0.000	425.43334
R44	4,968.254	558.57582	0.000	455.22379
R45	0.000	881.20241	0.000	673.71981
R46	0.000	583.88172	0.000	284.40026
R47	2,000.000	117.90825	0.000	227.74805
R48	0.000	401.77504	0.000	52.20508
R49	0.000	41.22391	0.000	122.24383
R50	0.000	372.11595	0.000	121.28373
R51	0.000	22.69494	0.000	118.57797
R52	0.000	1,557.46440	0.000	382.74471
R53	0.000	1,200.99025	0.000	490.82291
R54	0.000	180.00463	0.000	70.24819
R55	0.000	275.24786	0.000	116.16355
R56	0.000	343.28388	0.000	97.52983
R57	0.000	892.51347	0.000	911.04255
R58	6,335.520	1,265.83293	10,009.162	4,742.56475
R59	0.000	1,717.09820	0.000	253.33779
R60	0.000	116.42146	0.000	99.50167

9 Conclusion

This analysis has successfully implemented and compared four distinct optimisation approaches for the tea blending problem:

1. **Linear Programming** provides the mathematically optimal cost solution, serving as an essential

benchmark. However, it treats quality deviations as hard constraints within tolerance, not as objectives to minimise.

2. **Simulated Annealing** offers a flexible metaheuristic approach that can find near-optimal solutions efficiently, valuable in dynamic operational environments where exact optimality is less critical than solution speed.
3. **Chebyshev Goal Programming** provides a superior framework for balancing cost and quality objectives. By formally incorporating decision-maker preferences through goals and weights, it delivers balanced, actionable solutions that directly manage the trade-off between production cost and product quality.
4. **Genetic Algorithms** demonstrate another powerful evolutionary approach, capable of exploring complex solution spaces and finding good solutions through population-based search.

The comparative analysis reveals that while LP provides the lowest cost, multi-objective approaches like Chebyshev Goal Programming are better aligned with business realities where maintaining brand integrity through consistent quality characteristics is as important as minimising cost.

10 Appendix: Code Execution Notes

10.1 Required Libraries

All required R packages can be installed using:

```
install.packages(c("dplyr", "Rglpk", "ompr", "ompr.roi", "ROI", "ROI.plugin.glpk",  
                  "lpSolveAPI", "ggplot2", "GA", "knitr", "flextable", "tidyr"))
```

10.2 Reproducibility

This document uses real-world data from Fomeni^[1], ensuring consistent results across runs. Stochastic algorithms (Simulated Annealing and Genetic Algorithms) may produce slightly different results on each run due to their probabilistic nature, but the underlying data and deterministic algorithms (Linear Programming and Goal Programming) will produce identical results.

10.3 Parameter Tuning

Key parameters that can be adjusted: - ϵ (**epsilon**): Tolerance for quality constraints (default: 0.5 for Fomeni data) - **SA parameters**: Initial temperature, cooling rate, iterations - **GA parameters**: Population size, generations, mutation/crossover rates - **Goal Programming weights**: Cost vs quality priorities

11 Appendix: Complete Recipe Matrices

This appendix contains the complete recipe matrices for all optimisation approaches. Each matrix shows the amount (in kg) of each raw material used in each blend.

11.1 Linear Programming Solution

```
## === LP Solution: Complete Recipe Matrix (x_ij in Kgs) ===
```

Table 13: LP Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R2	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	1,000.000	0
R3	0.000	0.00000	20,450.410	0.000	4,549.590	0.0000	0.0000	0.0000	0.000	0
R4	0.000	0.00000	0.000	0.000	1,000.000	0.0000	0.0000	0.0000	0.000	0
R5	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R6	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	6,333.333	0
R7	0.000	0.00000	0.000	20,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R8	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	1,000.000	0
R9	0.000	0.00000	2,207.300	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R10	0.000	0.00000	0.000	0.000	0.000	15,000.0000	0.0000	0.0000	0.000	0
R11	40,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R12	7,554.827	50,277.77778	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R13	0.000	0.00000	0.000	4,164.021	0.000	0.0000	0.0000	0.0000	0.000	0
R14	5,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R15	0.000	0.00000	0.000	0.000	0.000	0.0000	2,333.3333	2,666.6667	0.000	0
R16	0.000	0.00000	0.000	0.000	1,250.000	0.0000	0.0000	0.0000	0.000	18,750
R17	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	833.3333	0.000	0
R18	0.000	20,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R19	0.000	3,746.03175	0.000	16,253.968	0.000	0.0000	0.0000	0.0000	0.000	0
R20	4,371.720	0.00000	11,543.614	0.000	24,084.666	0.0000	0.0000	0.0000	0.000	0
R21	0.000	0.00000	0.000	0.000	13,384.326	0.0000	0.0000	0.0000	0.000	0
R22	0.000	0.00000	0.000	19,666.667	0.000	333.3333	0.0000	0.0000	0.000	0
R23	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R24	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R25	0.000	0.00000	0.000	500.000	0.000	0.0000	0.0000	0.0000	0.000	0
R26	0.000	0.00000	1,000.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R27	0.000	0.00000	19,170.041	6,082.011	14,747.948	0.0000	0.0000	0.0000	0.000	0
R28	0.000	0.00000	0.000	7,333.333	0.000	32,666.6667	0.0000	0.0000	0.000	0
R29	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R30	0.000	0.00000	0.000	0.000	0.000	0.0000	338.9831	0.0000	0.000	0
R31	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R32	0.000	0.00000	3,333.333	0.000	0.000	0.0000	0.0000	0.0000	1,666.667	0
R33	0.000	0.00000	0.000	0.000	0.000	0.0000	4,615.8192	0.0000	0.000	0

Table 13: LP Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R34	0.000	0.00000	0.000	0.000	9,500.000	0.0000	0.0000	500.0000	0.000	0
R35	8,769.679	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R36	0.000	7,523.80952	0.000	0.000	0.000	0.0000	2,118.6441	0.0000	0.000	0
R37	0.000	0.00000	0.000	10,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R38	20,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R39	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R40	0.000	0.00000	0.000	0.000	2,000.000	0.0000	0.0000	0.0000	0.000	0
R41	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R42	1,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R43	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R44	4,968.254	31.74603	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R45	0.000	0.00000	0.000	0.000	1,489.625	0.0000	0.0000	0.0000	0.000	0
R46	0.000	0.00000	0.000	0.000	0.000	2,000.0000	0.0000	0.0000	0.000	0
R47	2,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R48	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R49	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R50	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R51	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R52	0.000	0.00000	0.000	0.000	5,000.000	0.0000	0.0000	0.0000	0.000	0
R53	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R54	0.000	0.00000	0.000	0.000	0.000	0.0000	593.2203	0.0000	0.000	0
R55	0.000	0.00000	0.000	1,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R56	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	1,000.0000	0.000	0
R57	0.000	0.00000	2,295.301	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R58	6,335.520	9,420.63492	0.000	0.000	22,993.845	0.0000	0.0000	0.0000	0.000	1,250
R59	0.000	5,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R60	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0

11.2 Simulated Annealing Solution

=== SA Solution: Complete Recipe Matrix (x_{ij} in Kgs) ===

Table 14: SA Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R1	1,331.85523	577.78237	745.407697	557.657449	1,160.225293	354.75090	9.529329	7.7298461	30.395441
R2	118.25196	265.97820	14.798911	186.539669	232.665092	75.53110	1.026690	1.8105726	11.870389
R3	3,229.15059	4,008.14311	3,019.294008	5,204.395532	2,521.335377	820.82880	597.177842	184.9346581	127.064381
R4	265.63224	58.20129	41.246842	233.041558	217.914739	41.14273	33.442545	2.3829763	19.988471
R5	4,171.30868	1,200.17949	2,290.042303	1,372.633178	5,545.211464	2,017.76220	619.780057	37.5118734	316.236402
R6	5,109.09689	1,046.69403	1,369.190198	135.658761	4,520.845848	259.88025	108.455815	108.0588571	350.446055
R7	4,888.28021	3,599.16530	2,080.565779	924.892395	4,320.304772	802.77422	665.385369	136.1202742	95.346938
R8	70.95111	99.21051	101.067537	310.840274	281.559297	34.90492	34.297501	13.0531251	10.867584
R9	1,930.64596	3,682.85705	1,735.116050	2,379.499359	2,978.754585	1,156.19253	274.293608	160.6867228	508.143372
R10	2,153.39220	2,644.94403	1,993.105099	2,480.635488	2,925.306161	1,747.39336	202.357090	137.0992138	13.337817
R11	828.26122	3,115.10290	545.405857	4,502.484275	2,280.627504	363.20706	163.071965	155.9174370	557.166138
R12	7,108.47289	2,406.32025	2,473.624074	3,290.032073	5,171.117373	466.29650	166.321051	167.4150103	87.559099
R13	5,408.09618	3,277.43682	2,142.708678	3,226.121004	1,528.479877	259.16346	647.712684	179.3611226	387.864186
R14	1,153.62356	1,027.98293	428.843447	832.047883	555.822833	447.84127	130.231566	12.9680779	190.872354
R15	520.77789	469.60796	1,045.308580	1,351.742729	852.539996	110.78451	4.856881	87.1099480	124.042455
R16	4,844.67525	3,062.86773	1,935.827028	2,535.640229	2,566.204244	1,050.83174	79.087701	227.9143149	85.857176
R17	2,747.51850	1,599.46553	1,328.604037	4,309.689863	4,375.185674	1,962.12858	443.577697	195.2078367	241.633297
R18	4,776.27980	1,558.00605	2,953.715089	233.298456	5,777.900724	2,242.03008	1.193977	237.5498494	93.443875
R19	84.78120	5,648.34460	843.188311	5,836.232455	3,574.754193	2,785.17813	291.397665	168.9610630	159.641107
R20	2,435.43507	3,698.22335	4,342.443659	394.282709	2,392.481103	450.96650	443.554895	48.6768721	294.485465
R21	5,018.19509	6,120.19254	2,360.637567	4,400.241426	544.204645	2,120.29317	451.422365	125.5837244	514.626965
R22	1,995.82237	5,612.57580	569.772781	3,631.501058	6,046.863143	1,862.11345	113.839160	36.6402983	45.126544
R23	1,405.25696	265.87870	13.807269	1,643.911049	360.363514	565.22630	96.518880	44.1863149	149.848035
R24	232.96545	272.79580	73.145532	158.488405	26.353786	137.90098	11.451327	9.0062718	38.341988
R25	57.22893	75.35507	9.325365	211.050719	19.760126	83.38928	16.835984	2.9688548	19.438527
R26	126.97315	207.41699	248.895093	73.592128	90.444080	176.39648	0.000000	3.4977319	3.174670
R27	5,101.26631	3,523.59760	327.428770	6,916.295666	1,610.878931	2,110.81996	494.769525	257.2743268	16.585733
R28	1,193.48229	1,465.23112	3,981.021224	3,859.691465	706.595273	2,334.43718	413.094312	102.7933450	628.866916
R29	1,925.78794	5,602.30145	3,255.074939	6,097.705466	3,595.301173	1,083.42605	205.305514	245.6397328	580.368168
R30	3,051.27762	4,880.96679	2,823.939606	1,352.135322	6,047.115778	1,671.77124	329.283919	179.3862804	675.385104
R31	71.74027	518.59607	1,305.895062	972.188136	131.117250	1,182.39081	142.803216	13.9680341	224.756151
R32	325.98550	1,723.67196	384.517532	1,391.921708	311.613551	531.02907	47.056109	37.8404779	168.400268
R33	2,597.58688	3,020.85435	913.959327	1,194.405401	41.682049	1,406.19566	256.456347	95.1578922	61.330213

Table 14: SA Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R34	39.74955	2,628.48159	535.441324	833.742898	2,321.871772	1,749.35581	450.968214	118.2422489	387.385294
R35	2,339.90929	44.01448	864.267634	3,287.295787	1,153.238659	1,409.25406	320.629197	128.5826810	16.991613
R36	1,222.14316	2,240.43873	375.927288	1,402.821013	2,889.085868	1,005.75903	38.569080	40.9930427	97.557298
R37	1,939.95505	710.41945	1,424.027574	3,051.943786	753.273734	1,212.22798	284.001691	154.2210944	143.425140
R38	4,507.38889	3,496.20815	2,609.780076	4,596.380215	1,511.894929	1,930.40574	146.436833	116.4887568	343.856847
R39	813.50522	59.65353	1,968.865676	3,190.610978	5,796.891821	1,218.58126	249.112265	244.4298306	663.444311
R40	272.06474	477.17051	315.997708	268.316177	281.164860	186.89884	42.687233	19.7684355	51.351566
R41	942.28110	955.18813	438.787978	20.015746	2,008.965552	289.22541	27.774326	25.7563783	239.333846
R42	370.95680	193.91602	192.358011	44.591855	3.934465	66.81862	29.643096	9.9946110	41.278246
R43	743.75571	1,688.64478	231.337645	196.527398	1,121.026350	669.62624	23.243458	72.5065435	87.709897
R44	558.57582	1,292.83515	506.476764	1,518.979196	555.815427	74.62490	199.795985	67.2898879	165.221559
R45	881.20241	815.72817	139.061678	909.942330	1,466.313876	665.58628	18.534230	31.4849326	39.959611
R46	583.88172	440.97130	147.495917	20.549427	203.778655	274.91544	10.864700	17.6067654	23.042467
R47	117.90825	214.90434	101.279296	322.217506	778.557700	327.98357	13.262698	30.9954000	3.609724
R48	401.77504	62.85990	6.929096	42.901275	275.503389	128.28850	34.125419	0.8576497	19.366949
R49	41.22391	69.36355	45.900275	288.124297	262.400851	125.10171	24.269292	4.5461541	28.544398
R50	372.11595	112.15694	149.747460	5.923281	229.171980	16.40431	34.380801	3.1671019	38.404948
R51	22.69494	454.07630	60.021897	246.657033	12.481223	79.00571	56.632467	1.4759008	9.093710
R52	1,557.46440	746.94732	403.895412	1,485.619692	10.534055	393.09621	104.913111	50.2185216	202.638982
R53	1,200.99025	1,276.21345	234.830383	534.189161	123.730620	814.03868	34.307530	76.3733086	37.576348
R54	180.00463	239.38673	85.242354	27.814484	286.919917	82.12386	1.997228	11.9315431	39.804504
R55	275.24786	80.00834	142.151547	169.249629	190.072132	40.25889	19.556437	14.2259763	25.588979
R56	343.28388	35.07482	79.939571	33.375501	387.753776	50.49982	11.007038	3.0515373	38.495277
R57	892.51347	1,053.64373	624.700706	949.064829	607.741459	419.25681	117.724646	10.6196101	97.725109
R58	1,265.83293	4,094.50768	631.367850	3,576.496802	2,578.428498	2,781.28042	108.319992	242.6359987	328.287050
R59	1,717.09820	170.81109	350.580466	389.321349	908.378735	1,115.21253	69.728334	74.8259707	73.870379
R60	116.42146	10.42811	83.525899	386.835064	61.187254	159.19090	31.926112	7.3392604	7.034892

11.3 Chebyshev Goal Programming Solution

=== Chebyshev GP Solution: Complete Recipe Matrix (x_{ij} in Kgs) ===

Table 15: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R2	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R3	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R4	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R5	0.000	0.0000	1,234.6787	0.0000	0.0000	0.000	0	0	0	0.0000
R6	0.000	0.0000	7,958.0429	0.0000	12,041.9571	0.000	0	0	0	0.0000
R7	0.000	0.0000	8,359.9383	0.0000	0.0000	1,890.062	0	0	9,750	0.0000
R8	0.000	0.0000	0.0000	0.0000	0.0000	1,000.000	0	0	0	0.0000
R9	0.000	102.3406	0.0000	14,897.6594	0.0000	0.000	0	0	0	0.0000
R10	0.000	0.0000	0.0000	15,000.0000	0.0000	0.000	0	0	0	0.0000
R11	40,000.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R12	32,528.357	38,998.6023	0.0000	0.0000	8,473.0411	0.000	0	0	0	0.0000
R13	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	1,100	0	0.0000
R14	0.000	0.0000	0.0000	0.0000	0.0000	0.000	5,000	0	0	0.0000
R15	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R16	0.000	1,231.6725	0.0000	0.0000	0.0000	0.000	0	0	0	18,768.3275
R17	0.000	0.0000	0.0000	0.0000	0.0000	3,082.454	3,000	3,900	0	0.0000
R18	3,328.753	0.0000	0.0000	0.0000	16,671.2474	0.000	0	0	0	0.0000
R19	0.000	0.0000	0.0000	0.0000	20,000.0000	0.000	0	0	0	0.0000
R20	0.000	0.0000	0.0000	40,000.0000	0.0000	0.000	0	0	0	0.0000
R21	0.000	18,417.3750	18,134.9499	3,447.6751	0.0000	0.000	0	0	0	0.0000
R22	0.000	0.0000	0.0000	7,615.9712	7,509.4432	0.000	0	0	0	0.0000
R23	0.000	0.0000	0.0000	5,000.0000	0.0000	0.000	0	0	0	0.0000
R24	0.000	0.0000	0.0000	0.0000	516.0415	0.000	0	0	0	0.0000
R25	0.000	0.0000	0.0000	0.0000	500.0000	0.000	0	0	0	0.0000
R26	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R27	0.000	0.0000	21,460.7884	0.0000	0.0000	14,027.485	0	0	0	0.0000
R28	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R29	0.000	0.0000	0.0000	0.0000	0.0000	25,000.000	0	0	0	0.0000
R30	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R31	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R32	0.000	0.0000	0.0000	0.0000	0.0000	5,000.000	0	0	0	0.0000
R33	0.000	0.0000	0.0000	0.0000	0.0000	0.000	2,000	0	0	0.0000

Table 15: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R34	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R35	0.000	0.0000	0.0000	0.0000	624.2942	0.000	0	0	0	0.0000
R36	0.000	4,571.0624	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R37	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R38	14,133.729	0.0000	0.0000	1,902.9377	0.0000	0.000	0	0	0	0.0000
R39	0.000	0.0000	0.0000	0.0000	15,000.0000	0.000	0	0	0	0.0000
R40	0.000	2,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R41	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R42	0.000	0.0000	339.4425	0.0000	0.0000	0.000	0	0	250	410.5575
R43	0.000	0.0000	0.0000	0.0000	5,000.0000	0.000	0	0	0	0.0000
R44	0.000	0.0000	0.0000	522.9126	4,477.0874	0.000	0	0	0	0.0000
R45	0.000	0.0000	0.0000	0.0000	5,000.0000	0.000	0	0	0	0.0000
R46	0.000	0.0000	0.0000	2,000.0000	0.0000	0.000	0	0	0	0.0000
R47	0.000	0.0000	0.0000	0.0000	2,000.0000	0.000	0	0	0	0.0000
R48	0.000	1,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R49	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R50	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R51	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	821.1150
R52	0.000	2,487.8407	2,512.1593	0.0000	0.0000	0.000	0	0	0	0.0000
R53	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R54	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R55	0.000	529.7838	0.0000	470.2162	0.0000	0.000	0	0	0	0.0000
R56	0.000	754.9784	0.0000	245.0216	0.0000	0.000	0	0	0	0.0000
R57	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R58	10,009.162	28,906.3443	0.0000	0.0000	1,084.4942	0.000	0	0	0	0.0000
R59	0.000	0.0000	0.0000	3,897.6061	1,102.3939	0.000	0	0	0	0.0000
R60	0.000	1,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000

11.4 Genetic Algorithm Solution

=== GA Solution: Complete Recipe Matrix (x_{ij} in Kgs) ===

Table 16: GA Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R1	545.81891	648.71545	397.54080	387.85215	725.23411	477.29785	241.93371	565.65756	476.064
R2	57.61176	87.08950	92.25326	45.75834	132.80467	142.06863	134.63848	113.88332	61.091
R3	3,621.03608	2,143.43294	2,042.30911	1,861.88471	3,718.85189	3,107.34922	3,705.54532	2,071.62112	1,565.692
R4	89.17773	108.91474	98.89243	100.34063	84.40016	115.33216	89.72693	171.36910	86.234
R5	2,626.83581	3,131.94732	1,813.93694	1,980.48302	1,915.47490	3,636.38860	2,490.06248	3,137.11705	2,782.740
R6	2,568.15947	2,856.21692	2,166.78017	1,664.87607	1,763.68935	1,806.83770	1,765.05802	2,800.51897	1,938.504
R7	3,052.05712	1,781.78000	1,032.59946	3,097.25110	2,182.14717	1,913.41683	2,348.38917	1,766.27178	2,041.453
R8	75.81071	119.86658	117.98637	106.05478	84.79108	114.60933	119.04524	56.32557	102.541
R9	1,265.52356	1,967.85617	1,561.26464	1,890.39893	1,787.93201	1,146.10502	1,747.80720	1,371.10366	1,589.394
R10	1,517.95210	1,348.44885	1,583.05893	1,174.49679	2,081.34314	2,655.15776	820.34994	1,008.02417	1,847.716
R11	3,334.23571	6,257.98417	589.72873	6,970.46849	6,570.45972	3,292.33851	2,832.48225	4,203.37900	2,435.782
R12	12,743.72460	11,101.31102	3,824.95496	13,653.84177	10,422.50861	2,925.34314	5,207.82348	8,545.56718	10,324.752
R13	7,195.17407	7,250.63440	5,153.71967	5,794.98697	7,340.32126	3,886.30787	7,377.60198	6,489.76740	5,412.622
R14	685.94362	412.83462	238.34292	737.41315	723.07704	696.08259	617.50850	257.91627	458.307
R15	741.35734	650.40155	412.80166	531.16065	456.06816	224.37543	323.63396	468.21491	497.751
R16	1,504.90084	3,658.35329	2,515.41448	2,223.85238	1,448.70341	660.25914	1,588.39955	1,840.51100	2,972.172
R17	2,902.41474	2,021.81333	2,367.96770	1,653.52124	1,521.27361	2,485.19283	2,321.55401	1,419.98136	777.158
R18	2,121.22076	2,589.78736	2,275.23877	2,465.94911	2,327.54235	1,924.73046	2,681.06451	1,707.53138	1,028.090
R19	1,240.75570	2,612.52628	1,960.08566	2,360.33493	1,634.52812	3,322.50475	2,556.51383	1,765.24594	1,233.852
R20	6,801.31444	5,097.46954	4,330.57731	3,456.54031	3,124.63189	4,074.22175	3,982.83025	3,106.73523	4,219.549
R21	4,616.45083	8,067.81967	1,835.75043	5,739.76659	5,577.03120	2,961.91049	1,948.11477	4,592.40730	2,552.453
R22	1,645.34408	2,449.46431	1,212.26355	2,478.31508	3,232.33775	2,262.47272	2,486.94550	1,308.60260	726.510
R23	757.77331	337.38767	612.22972	399.34143	658.17411	225.36717	634.44609	299.01951	378.907
R24	100.45735	84.94428	80.91781	148.42405	89.90171	47.78020	142.10694	120.93736	88.308
R25	23.92929	48.54299	65.19982	46.30279	48.68188	41.08489	58.05422	45.62702	67.116
R26	91.69086	127.54646	119.48247	93.02806	91.87369	139.08649	44.05254	50.58146	137.453
R27	5,087.12299	3,351.03348	3,211.24433	6,666.86298	3,518.78320	6,588.48563	4,130.74720	970.71612	5,608.121
R28	4,829.44380	5,221.54164	3,094.82896	5,892.20936	5,957.17300	3,404.81695	3,133.62805	3,942.41799	2,683.203
R29	3,103.53837	1,746.56894	1,785.39380	3,308.26121	3,732.45643	2,351.68891	1,552.75695	2,022.78192	3,882.615
R30	3,349.97226	2,540.15755	2,310.26732	3,798.08901	5,090.76795	3,347.27124	1,540.18287	1,433.42890	4,321.544
R31	491.01632	733.10694	442.69458	702.88314	513.64423	537.84795	521.27865	636.67820	328.693
R32	436.23961	394.79226	622.30988	284.22482	718.62087	650.83751	646.56711	229.06338	423.990
R33	1,098.96626	1,160.15317	789.33438	1,258.87184	792.09468	1,233.07329	979.36496	919.72916	555.086

Table 16: GA Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R34	1,224.76929	909.32754	464.74637	1,034.96453	1,049.39836	1,251.50897	952.60062	1,356.53827	742.387	742.387
R35	1,033.00212	1,126.66065	267.62695	764.20476	1,495.04268	1,677.27839	948.01095	466.07524	756.911	756.911
R36	1,207.41370	533.71905	375.36541	869.43020	1,280.52450	1,063.39634	1,422.14992	1,245.17563	1,183.765	1,183.765
R37	1,301.90768	1,280.41869	1,149.12507	920.62763	1,566.48379	969.05070	963.95203	747.72557	503.432	503.432
R38	3,549.94343	2,274.60643	1,510.83284	1,844.96285	2,016.98180	2,174.97231	3,036.82782	587.21861	1,610.924	1,610.924
R39	949.95851	1,314.31287	1,155.29468	1,326.79300	1,770.45608	2,115.85594	1,408.36497	1,601.37898	1,448.974	1,448.974
R40	168.33999	184.85506	158.01081	235.65366	158.38787	181.64936	150.75122	211.97468	299.112	299.112
R41	432.53532	596.20775	326.90866	416.89062	426.18941	807.77032	508.20982	524.06544	641.453	641.453
R42	164.36775	102.41002	83.45797	49.01429	85.73714	84.01294	153.29566	105.13870	58.679	58.679
R43	425.43334	738.13283	497.45501	502.93286	517.45261	437.08102	555.46446	741.11435	234.571	234.571
R44	455.22379	594.34097	540.86949	344.25182	475.61389	493.58689	738.44886	581.56971	449.966	449.966
R45	673.71981	687.10531	389.71655	800.23406	377.32284	397.16973	314.88046	656.16230	470.881	470.881
R46	284.40026	246.35741	157.42124	221.55305	139.23771	108.31563	206.85186	271.93788	143.152	143.152
R47	227.74805	255.89722	169.81261	263.23817	166.79470	186.44484	218.98731	206.62266	118.468	118.468
R48	52.20508	96.93189	107.70826	78.10258	145.37779	111.04339	61.37236	55.17266	156.486	156.486
R49	122.24383	113.03949	110.21829	114.49246	96.10615	66.85001	127.06745	95.15238	84.630	84.630
R50	121.28373	61.86328	77.94448	103.15495	131.65991	72.42718	107.88987	107.02983	132.953	132.953
R51	118.57797	139.19575	110.67324	94.83619	48.63825	100.09545	83.21715	40.94691	174.219	174.219
R52	382.74471	409.85228	465.19299	500.75779	446.00308	588.66026	794.64664	175.88038	661.306	661.306
R53	490.82291	464.11990	360.73234	524.19508	671.01302	360.85752	531.47328	565.91705	651.442	651.442
R54	70.24819	96.04305	120.84196	126.72525	87.75845	43.24437	82.28018	76.38469	155.772	155.772
R55	116.16355	132.77039	75.86966	91.47886	176.27806	39.72741	99.06272	101.19512	42.861	42.861
R56	97.52983	123.19024	96.85471	112.02304	129.45972	120.06319	52.06943	63.18677	123.352	123.352
R57	911.04255	553.39385	524.49550	539.77307	657.74750	356.60746	183.80746	312.12568	427.679	427.679
R58	4,742.56475	4,246.99102	4,972.87684	4,251.33120	4,955.64113	3,990.03346	2,154.30947	1,936.05149	3,965.222	3,965.222
R59	253.33779	485.51388	466.59190	780.12179	787.42979	475.44793	593.45152	588.42938	252.697	252.697
R60	99.50167	122.29977	123.67277	114.21039	73.94041	74.62363	61.15287	77.42792	112.507	112.507

References

1. Djeumou Fomeni, F. (2018). A multi-objective optimization approach for the blending problem in the tea industry. *International Journal of Production Economics*, 205, 179–192. <https://doi.org/10.1016/j.ijpe.2018.08.036>